INCH-POUND
J-W-1177/32
June 10, 1988

FEDERAL SPECIFICATION SHEET

WIRE, MAGNET, ELECTRICAL, CLASS 180, TYPE DgO POLYESTER-GLASS-FIBER-COVERED, ORGANIC VARNISH TREATED, ROUND

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

The requirements for acquiring the wire described herein shall consist of this specification and the latest issue of J-W-1177.

Classification:	Class 180; type DgO and type Dg20 (bare with single or double polyester-glass fiber, varnished), type LDgO and type LDg20 (single film, single or double polyester-glass fiber, varnished), type L2DgO and type L2Dg20 (heavy film, single or double polyester-glass fiber, varnished); round.
Insulating materials:	The fiber covering and application of the covering shall be as specified in J-W-1177. If an underlying film coating is used, it shall have a class 155 rating. The varnish used in treating fibrous covered wire shall conform to the requirements of class 180 of MIL-I-24092, or an alternate selected on the basis of equivalent test data. The varnish shall be a high temperature non-silicone insulating varnish. The varnish used shall be identified in the qualification test report.
NEMA/ANSI equivalent:	All test requirements are equivalent to MW-51 of NEMA MW 1000.
General requirements:	See J-W-1177 for general requirements, quality assurance provisions, and packaging.

AMSC N/A FSC 6145

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Requirements:

Characteristics	Test procedure, see J-W-1177	Wire sizes, AWG	Requirements
Dimensions	4.7.1.2	4-30	See table I.
Adherence and flexibility	4.7.2.2.1	4-30	Covering shall not open sufficiently to expose the bare or underlying film-coated wire after bending 9 AWG and heavier wire on a 10X mandrel and 10-30 AWG wire on a 5X mandrel.
	4.7.2.2.2	4-30	There shall be no loosening, fraying or loss of adherence of the covering except at the point of rupture.
Elongation	4.7.5	4-30	Not less than the values shown in table II.
Dielectric strength	4.7.9	4-30	Not less than the values shown in table III.
Thermal endurance	----	4-30	Class 180. All insulating materials shall meet the thermal class ratings as described above.

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TABLE I. Dimensions.

AWG size	Bare wire diameter, inch			Minimum increase single polyester glass fiber covering, inch	Single covering			Minimum increase single polyester glass fiber covering, inch	Double covering		
					Maximum overall diameter, inch				Maximum overall diameter, inch		
	Minimum	Nominal	Maximum		Bare	Single film	Heavy film		Bare	Single film	Heavy film
4	0.2023	0.2043	1/ 0.2053	0.0040	----	----	0.2158	0.0060	0.2153	----	0.2188
5	.1801	.1819	1/ .1828	.0040	----	----	.1932	.0060	.1927	----	.1962
6	.1604	.1620	1/ .1628	.0040	----	----	.1731	.0060	.1726	----	.1761
7	.1429	.1443	1/ .1450	.0040	----	----	.1551	.0060	.1547	----	.1581
8	.1272	.1285	1/ .1292	.0040	----	----	.1392	.0060	.1388	----	.1422
9	.1133	.1144	1/ .1150	.0040	----	----	.1249	.0060	.1245	----	.1279
10	.1009	.1019	1/ .1024	.0035	0.1079	----	.1111	.0055	.1109	----	.1141
11	.0898	.0907	1/ .0912	.0035	.0968	----	.0998	.0055	.0996	----	.1028
12	.0800	.0808	1/ .0812	.0035	.0866	----	.0897	.0055	.0895	----	.0927
13	.0713	.0720	1/ .0724	.0035	.0777	----	.0807	.0055	.0807	----	.0837
14	.0635	.0641	1/ .0644	.0035	.0697	0.0716	.0732	.0055	.0727	0.0746	.0762
15	.0565	.0571	1/ .0574	.0035	.0627	.0644	.0659	.0055	.0657	.0674	.0689
16	.0503	.0508	1/ .0511	.0035	.0563	.0581	.0595	.0055	.0593	.0611	.0625
17	.0448	.0453	1/ .0455	.0035	.0508	.0525	.0538	.0055	.0538	.0555	.0568
18	.0399	.0403	1/ .0405	.0035	.0457	.0474	.0487	.0055	.0487	.0504	.0517
19	.0355	.0359	1/ .0361	.0035	.0413	.0429	.0441	.0055	.0443	.0459	.0471
20	.0317	.0320	1/ .0322	.0035	.0373	.0389	.0401	.0055	.0403	.0419	.0431
21	.0282	.0285	1/ .0286	.0035	.0338	.0353	.0364	.0055	.0368	.0383	.0304
22	.0250	.0253	1/ .0254	.0035	.0306	.0320	.0331	.0055	.0336	.0350	.0361
23	.0224	.0226	1/ .0227	.0035	.0278	.0293	.0303	.0055	.0308	.0323	.0333
24	.0199	.0201	1/ .0202	.0025	.0253	.0267	.0277	.0045	.0268	.0282	.0292
25	.0177	.0179	1/ .0180	.0025	.0231	.0244	.0253	.0045	.0246	.0259	.0268
26	.0157	.0159	1/ .0160	.0025	.0211	.0223	.0232	.0045	.0226	.0238	.0247
27	.0141	.0142	.0143	.0025	.0193	.0206	.0214	.0045	.0208	.0221	.0229
28	.0125	.0126	.0127	.0025	.0177	.0190	.0197	.0045	.0192	.0205	.0212
29	.0112	.0113	.0114	.0025	.0164	.0176	.0183	.0045	.0179	.0191	.0198
30	.0099	.0100	.0101	.0025	.0151	.0162	.0169	.0045	.0166	.0177	.0164

1/ The maximum bare wire dimensions may be exceeded up to the NEMA/ANSI maximum bare wire limit, provided the minimum increase is maintained and the maximum overall diameter specified is not exceeded.

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TABLE II. Elongation.

AWG size	Minimum elongation, percent
4-8	30.0
9-15	20.0
16-21	15.0
22-30	10.0

TABLE III. Minimum breakdown voltages.

AWG size	Diameter of mandrel, inches	Minimum breakdown, volts 1/	
		Single covering	Double covering
4-9	——	150	270
10-23	1.00	360	540
24-30	0.25	225	400

1/ For polyester-glass fiber covered wire having an underlying film coating, add the minimum breakdown voltage for the film coated wire.

Part number: Magnet wire covered by this specification shall be defined by the following part numbering system. Example: M1177/32-06C029.

M1177/32-	06	C	029
Federal specification identifier	Two digit type code	Single letter conductor code	Three character size code

The following codes shall apply:

Type	Type code	Conductor	Conductor code
Dg0	01	Copper	C
Dg20	02	Aluminum	A
LDg0	03	Nickel-coated copper	N
LDg20	04	Silver-coated copper	S
L2Dg0	05		
L2Dg20	06		

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The size code shall be the bare wire dimensions. AWG wire size shall be used.

Intended use: Type DgO magnet wire is intended for use in 180°C applications similar to GO where increased toughness, flexibility and nonfraying properties are required.

MILITARY INTERESTS:

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - AR, ER, MI
DLA - IS

User activities:
Army - ME
Navy - AS, CG, MC, OS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS, PBO, PCD
INTERIOR - BLM
HHS - FDA
DCGOVT - DCG
NASA - JFK
COMMERCE - NBS
TRANSPORTATION - APM, FAA

Preparing activity:
Navy - SH
(Project 6145-1111-28)

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